# Practice Set 8 Paper 3 – Subspecialities of Urology

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*Practice paper compiled from the question bank.*

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## Q1. Describe normal development of bladder and ureter and outline management of a child born with ectopia vesicae.

**Type:** LAQ (25 marks) (bladder embryology and exstrophy theories also answered in Practice Set 6 Paper 1, Q2)

*Assumed meaning: ectopia vesicae = classic bladder exstrophy (CBE).*

### A. Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds → duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. Bud stalk → ureter; its branches → pelvis, calyces and collecting ducts; blastema → nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

### B. Development of the bladder

- **Cloacal division (weeks 5–6):** the **urorectal septum** divides the cloaca into dorsal **anorectal canal** and ventral **UGS (urogenital sinus)**.
- **Wolffian duct entry** is the landmark: cranial **vesicourethral canal** → bladder and pelvic urethra; caudal **definitive UGS** → phallic urethra (male), vestibule (female).
- **Layers:** urothelium from UGS endoderm; detrusor and adventitia from splanchnic mesenchyme by epithelial–mesenchymal signalling.
- **Urachus:** the bladder descends (months 4–5); the allantois obliterates to the urachus (median umbilical ligament).
- Urine from week 10; after week 16 amniotic fluid is mainly fetal urine.

### C. Trigone and ureterovesical junction

- **Weeks 5–6:** the **common excretory duct** (Wolffian duct distal to the bud) is absorbed into the UGS; **apoptosis** separates ureter from Wolffian duct. The arms rotate (**Mackie–Stephens**): ureteric orifice moves cranially and laterally; Wolffian duct ends caudally as the ejaculatory duct.
- **Trigone muscle:** classic view, ureteric extension; current lineage tracing, mostly bladder smooth muscle.

| Bud origin | Orifice | Result |
|---|---|---|
| Too low (early contact) | High, lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too high (late contact) | Caudal, medial | Ectopic ureter, often obstructed |
| Either | Abnormal bud–blastema contact | Dysplasia (VUR and dysplasia coexist) |

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

### D. Ectopia vesicae: basis and anatomy

- **Cause:** the cloacal membrane is not reinforced by mesoderm and ruptures (**Marshall–Muecke** overdevelopment theory). Timing decides cloacal exstrophy, CBE or epispadias.
- **Incidence:** about 1 in 46,000; male:female 2.3:1.
- **Defects:** exposed bladder plate; pubic diastasis; epispadias with short penis (or bifid clitoris); low umbilicus; anterior anus; hernias. **VUR in 100%** after closure. Bladder is immature, not neuropathic.

## E. Management

**1. Prenatal:** ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia, lower abdominal mass. Counsel; deliver at an exstrophy centre.

**2. At birth**

- Tie cord with silk (no clamp); cover plate with plastic wrap; saline irrigation at nappy change.
- Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template: delay closure, not a bad early closure.

**3. Pelvic osteotomy and immobilisation**

- **Indications:** diastasis **>4 cm**, stiff pelvis, closure **after 72 hours**, failed closure, cloacal exstrophy.
- **Combined bilateral anterior innominate + vertical iliac osteotomy** with external fixator and traction. Without osteotomy: modified Bryant traction 4 weeks. Spica casting fails more often.

**4. MSRE (modern staged repair of exstrophy)**

| Age | Stage | Purpose |
|---|---|---|
| Newborn | Bladder, posterior urethra and abdominal wall closure | Converts to complete epispadias; outlet resistance stimulates bladder growth |
| 6–12 months | Epispadias repair (modified Cantwell-Ransley), after testosterone | Raises capacity; straightens penis |
| 4–7 years | BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplant | Continence; needs capacity **≥100 mL** and a motivated child |

- **Closure keys:** radical division of the **urogenital diaphragm fibres** so the vesicourethral unit sits deep (commonest error if missed); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (complete primary repair of exstrophy, Mitchell), Kelly radical soft-tissue mobilisation, Erlangen single stage, Mainz ureterosigmoidostomy. Most still need an outlet procedure.

**5. Follow-up**

- Prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.

**6. Failures and salvage**

- Dehiscence or prolapse: wait 4–6 months, reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff**.

## F. Outcomes and long-term

- Day continence after MSRE about **70–75%**. **Best predictors: template size at birth and a successful first closure.**
- **Also know:** adenocarcinoma risk (cystoscopy; colonoscopy after ureterosigmoidostomy); uterine prolapse; reduced male fertility; psychosocial support.

**Diagram to draw:** Wolffian duct with ureteric bud and its absorption into the UGS; MSRE timeline.

**Key points to remember:**

- Bud from Wolffian duct; position decides VUR or ectopia.
- Osteotomy if diastasis >4 cm or closure after 72 h.
- MSRE: birth closure, epispadias 6–12 mo, BNR at ≥100 mL.
- Template size + first successful closure predict continence.

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## Q2. enumerate various therapeutic energy applications in urology and briefly give their present status in various urological diseases.

**Type:** LAQ (25 marks) (energy physics also answered in 2025-Winter Paper 4, Q2)

### Introduction

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

### Classification

| Group | Modalities |
|---|---|
| Electrical | Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration) |
| Electromagnetic | Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation |
| Acoustic | SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy) |
| Direct contact | Cryoablation, water vapour (Rezum), waterjet (Aquablation) |

### 1. Electrosurgery

- **Principle:** alternating current at about **500 kHz** heats tissue (the resistance) without nerve stimulation. **Current density** decides the effect. Cut = continuous low voltage; coagulation = interrupted high voltage.
- **Status:**
  - **TURP (transurethral resection of prostate):** still the reference BPH (benign prostatic hyperplasia) operation. **Bipolar** in saline matches monopolar efficacy and avoids TUR syndrome (glycine hyponatraemia).
  - **TURBT (transurethral resection of bladder tumour):** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.
  - Vessel sealers (up to 7 mm vessels) are routine in laparoscopic and robotic surgery.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

### 2. Lasers

| Laser | nm | Penetration | Present status |
|---|---|---|---|
| **Ho:YAG** (holmium:yttrium-aluminium-garnet) | 2120 | 0.31 mm | Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation |
| **TFL** (thulium fibre laser) | 1940 | 0.08 mm | Stones: beats low-power holmium, equals high-power; less retropulsion; ThuLEP (thulium laser enucleation of prostate) |

| Laser | nm | Penetration | Present status |
|---|---|---|---|
| **KTP/LBO** (potassium-titanyl-phosphate/lithium triborate; GreenLight) | 532 | 0.8 mm | Photovaporisation; can continue anticoagulation |
| Tm:YAG (thulium:YAG) | 2000 | 0.2 mm | Vaporesection, enucleation |
| Nd:YAG (neodymium:YAG) | 1064 | 10 mm | Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH |

- **Mechanism:** pulse duration decides: long pulses photothermal; short pulses photomechanical.

## 3. Acoustic energy

- **SWL:** shock waves break brittle stones by spallation, shear and cavitation. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones; slower rate (about 1 Hz) works better.
- **Intracorporeal:** pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam heats a rice-grain focus to necrosis. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy, burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

## 4. Thermal ablation of tumours

| Modality | Mechanism | Present status |
|---|---|---|
| **Cryoablation** | Argon expansion (Joule–Thomson); **<−40 °C, two freeze–thaw cycles**; visible ice ball | Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate |
| **RFA** | 375–900 kHz; target 60 °C; char at 150 °C | Small renal mass; oncological results similar to cryo; heat sink near vessels |
| **MWA** | 915 MHz or 2.45 GHz; dielectric hysteresis | Renal mass; hotter, less heat sink |
| **IRE** | Microsecond DC (direct current) pulses open permanent membrane pores; non-thermal | Focal prostate; good continence and potency; trials ongoing |

- **Guidelines:** AUA (American Urological Association) 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

## 5. Light and chemical energy

- **PDT:** photosensitiser + light → singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard.

## 6. Prostate (BPH) newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

## 7. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours). **Radioligand therapy** ($^{177}$Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

**Diagram to draw:** laser penetration depths compared, with cryoprobe ice ball.

**Key points to remember:**

- Holmium 2120 nm: gold standard for stones and HoLEP.
- Bipolar TURP in saline avoids TUR syndrome.
- Thermal ablation for cT1a ≤3 cm; cryo <−40 °C, two cycles.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

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# Q3a. Sertoli cell only syndrome

**Type:** SAQ (10 marks) (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

## Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

## Types

| Type | Feature |
|---|---|
| Pure (complete, congenital) | All tubules lack germ cells |
| Mixed (focal, acquired) | Some tubules keep foci of spermatogenesis, the basis of sperm retrieval |

## Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

## Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

## Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

| Pattern | Meaning | Micro-TESE retrieval |
|---|---|---|
| Hypospermatogenesis | All stages, fewer | Best |
| Maturation arrest | Stops at one stage | About 40% |
| **SCOS** | No germ cells | **About 30%** |

## Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

**Diagram to draw:** tubule cross-section: normal layered epithelium versus Sertoli cells only.

**Key points to remember:**

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

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## Q3b. Oxidative stress and male infertility

**Type:** SAQ (10 marks)

### Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

### Sources of ROS

| Source | Examples |
|---|---|
| **Leucocytes** (main) | Infection, prostatitis, epididymitis, leucocytospermia |
| **Immature sperm** | Excess residual cytoplasm |
| Varicocele | Heat and hypoxia |
| Lifestyle | **Smoking**, alcohol, obesity, heat |
| Others | Ageing, diabetes, pollutants, radiation |

### Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

### Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

### Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

### Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.
- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.

- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

**Diagram to draw:** balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
>
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

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## Q3c. Dhat syndrome

**Type:** SAQ (10 marks)

### Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

### Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

### Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

### Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

### Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

### Management

**1. Counselling (main)**

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

**2. Psychological therapy:** CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

**3. Drugs**

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

**4. Avoid harm**

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.

- Refer to psychiatry when depression is significant.

## Prognosis

- Good with counselling; drop-out is common.

**Diagram to draw:** flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
>
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

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## Q3d. Eponymous urethroplasty techniques and perineal urethrostomy

**Type:** SAQ (10 marks) (perineal urethrostomy also answered in 2021-Summer Paper 2, Q3c)

### Introduction

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.

### Eponymous techniques

| Eponym | Technique | Use |
|---|---|---|
| **Johanson** | Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later | LS (lichen sclerosus), failed hypospadias, infection |
| **Blandy** | Inverted-U scrotal or perineal skin flap into a ventral urethrotomy | Bulbar strictures; classic perineal urethrostomy (PU) |
| **Turner-Warwick** | Two-stage scrotal inlay; transpubic approach, omental wrap | Complex, posterior |
| **Webster** | Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting | PFUI (pelvic fracture urethral injury) |
| **Barbagli** | **Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora | Bulbar |
| **Asopa** | **Dorsal inlay** graft through a ventral urethrotomy, no mobilisation | Penile and bulbar |
| **Kulkarni** | One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination | Panurethral (LS) |
| **Orandi** | Ventral longitudinal penile skin island flap on dartos | Pendulous |
| **Quartey** | Showed skin island is a passenger on a fascial (dartos) flap | Basis of all genital skin flaps |
| **McAninch** | Circular fasciocutaneous penile skin flap | Long anterior strictures (up to 15 cm) |
| **Jordan** | Vessel-sparing EPA; ventral transverse island for fossa | Proximal bulbar; fossa navicularis |

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

### Perineal urethrostomy (PU)

- **Definition:** bulbar urethra opened onto perineal skin; the patient voids sitting. It **bypasses** a diseased anterior urethra.
- **Indications:** LS panurethral disease, failed urethroplasties, elderly or high-risk patients, after penectomy.
- **Techniques:**
  - **Classic (Blandy):** inverted-U perineal flap into the opened bulbar urethra.
  - **BMG PU:** dorsal urethrotomy to healthy mucosa; ventral urethra to skin; calibrated to **30 Fr**; BMG on corpora completes the dorsal wall, preventing circumferential scarring.
- **Complications:** stenosis (main), spraying, perineal wetness.

**Diagram to draw:** bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

**Key points to remember:**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU; BMG PU at 30 Fr.

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# Q3e. Prognostic index score in carcinoma penis

**Type:** SAQ (10 marks)

*Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.*

## Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

## 1. Nodal status (most important factor)

| Nodal finding | 5-year survival |
|---|---|
| Node negative | About 73% |
| ≤2 nodes, unilateral, no ENE (extranodal extension) | About 80% |
| >2 nodes | 0–50% |
| **ENE** or **pelvic nodes** | Poor (pelvic about 14%) |

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

## 2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

## 3. Risk-group systems

| System | Low | Intermediate | High |
|---|---|---|---|
| **Earlier EAU (European Association of Urology), after Solsona** | Tis, Ta, T1 G1 | T1 G2 | T2 or more, or G3 |
| **Current EAU–ASCO (American Society of Clinical Oncology) / TNM** | PeIN, Ta, **T1a** (nodes 0–18%) | — | **T1b or more** (one-third to over half) |

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

## 4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).
- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

## 5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).
- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

## 6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

**Diagram to draw:** flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
>
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

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# Q3f. Paquin and Gil-Vernet techniques of ureteric reimplantation

**Type:** SAQ (10 marks)

## Principles of antireflux surgery

- Normal flap valve: intramural ureter compressed against detrusor during filling.
- **Paquin's rule:** tunnel length to ureteric diameter about **5:1** in non-refluxing junctions versus 1.4:1 in refluxing ones.
- Exclude secondary VUR (vesicoureteral reflux); tension-free, straight course.

## Indications for surgery

- Breakthrough febrile UTI (urinary tract infection) on prophylaxis, new scars, persistent high-grade VUR.

## 1. Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- **Steps:**
  1. Ureter mobilised **extravesically** and divided at the bladder.
  2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**).
  3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter.
  4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor.
  5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

## 2. Gil-Vernet trigonoplasty

- **Intravesical, no tunnelling and no ureteric mobilisation.**
- **Steps:**
  1. Bladder opened.
  2. **Transverse mucosal incision** across the trigone between the two orifices.
  3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline.
  4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters, paraureteral diverticulum.

## Comparison

| Feature | Paquin | Gil-Vernet |
|---|---|---|
| Approach | Extra + intravesical | Intravesical |
| Ureter divided | Yes | No |
| Tunnel | New long submucosal | None (advancement) |
| Best for | High grade, dilated | Low grade, bilateral |

- **Also know:** Politano–Leadbetter, Cohen (commonest, >98% success), Glenn–Anderson, Lich–Gregoir. Open success benchmark 95.9% (AUA [American Urological Association] meta-analysis).

**Diagram to draw:** trigone showing Gil-Vernet transverse incision and medial advancement stitch.

> **Key points to remember:**
> - Paquin: tunnel 5:1; combined, suprahiatal reimplant.
> - Gil-Vernet: trigonoplasty, medial advancement, no tunnel.
> - Gil-Vernet for low-grade bilateral VUR.

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# Rapid Revision Sheet – Practice Set 8 Paper 3

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Describe normal development of bladder and ureter and outline management of a child born with ectopia vesicae. (LAQ)**

- Bud from Wolffian duct; position decides VUR or ectopia.
- Osteotomy if diastasis >4 cm or closure after 72 h.
- MSRE: birth closure, epispadias 6–12 mo, BNR at ≥100 mL.
- Template size + first successful closure predict continence.

**Q2. enumerate various therapeutic energy applications in urology and briefly give their present status in various urological diseases. (LAQ)**

- Holmium 2120 nm: gold standard for stones and HoLEP.
- Bipolar TURP in saline avoids TUR syndrome.
- Thermal ablation for cT1a ≤3 cm; cryo <−40 °C, two cycles.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

**Q3a. Sertoli cell only syndrome (SAQ)**

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

**Q3b. Oxidative stress and male infertility (SAQ)**

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

**Q3c. Dhat syndrome (SAQ)**

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity.

**Q3d. Eponymous urethroplasty techniques and perineal urethrostomy (SAQ)**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU; BMG PU at 30 Fr.

**Q3e. Prognostic index score in carcinoma penis (SAQ)**

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

**Q3f. Paquin and Gil-Vernet techniques of ureteric reimplantation (SAQ)**

- Paquin: tunnel 5:1; combined, suprahiatal reimplant.
- Gil-Vernet: trigonoplasty, medial advancement, no tunnel.
- Gil-Vernet for low-grade bilateral VUR.